Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Team TEXTURA

Weekly Project Progress Report – Week 6

Course: Digital Image Processing

Repository link:

https://github.com/AditiKapadiya/DIP_2026_Project17_Group8.git

Team Members

Aditi Kapadiya	AU2440139
Disha Gadhiya	AU2440217
Khushi Panchal	AU2440002
Harshita Dhanwani	AU2440083

1. Work Done

Data and splits

Installed 3222 defect-source patches and 4371 defect-free patches. Of the defect patches, 367 contain real defect pixels (from 104 source images). There are 7 fabrics (00-06) with both kinds.

Defect patches were split DEV / TEST based on source image. Defect-free patches were divided by fabric into REF (learn the spectrum), CAL (calibrate thresholds) and NTEST (measure false alarms) also by source image to ensure no leakage between roles.

Spectral toolkit

Windowed FFT (Hann window) and detrending per window.

Peak detection in the log-spectrum (robust-sigma prominence), dilation of the notch mask and suppression of the lowest frequencies.

Overlapping windows, FFT, notch, inverse FFT, overlap-add of squared residuals, Gaussian smoothing, log scale (residual-energy engine)

Feasibility check

Built mean reference spectra (32/64/128/256 size fabrics).

All fabrics exhibit very strong periodic weaves with the dominant period of 4-6 px, and peak strength of z = 55-72. This notch subtracts 59-95% of the energy in the spectrum (highest notch = fabric 03, lowest notch = fabric 05).

Three pipelines compared

Used 128 scale (fixed) and multi-scale (fixed) (64/128/256) with 1 pooled reference and 1 smoothing for all fabrics.

Adaptive multi-scale: window sizes and smoothing set according to the estimated weave period of each fabric (at least 4 periods/window and about 1 period/smoothing).

Calibration and confidence

A false positive rate of 5% was used to define detection thresholds, calibrated on defect-free CAL patches (Gumbel fit to patch maxima). Fitted thresholds were from 4.38 to 5.56.

So, the probability of getting a lower score than the observed peak in a normal patch is the same as the confidence.

Mask generation

Ran four experiments: Per image Otsu, calibrated threshold, morphology clean-up, and morphology clean-up with hysteresis region growing. The growth fraction was adjusted for DEV (chosen: 0.7).

Confidence was built per-blob, overlays (detections and failure cases) were built, and a demo was built using the same CLAHE preprocessing and 256×256 patches with stride 128 was produced.

Results (DEV: 152 defect patches; NTEST: 280 normal patches)

Pipeline	**AP**	**Macro AP**	**AUROC**
Fixed single-scale	0.035	0.116	0.551
Fixed multi-scale	0.036	0.209	0.552
Adaptive multi-scale	0.080	0.084	0.797

Chance AP is 0.033 (the defect pixel fraction). Fixed multi-scale was carried forward because it had the highest Macro AP.

Final pipeline metric	**Value**
Defect patches detected	25.7% (39 of 152)

False alarms on normal patches	1.8% (5 of 280)
Image-level AUROC (confidence)	0.822
Mean IoU on detected patches	0.124

Confidence validation

- Observed false alarm shares at each confidence level are at or below expected levels (conservative calibration). For instance, the confidence scores of the above 0.90 patches are below 0.90 in 1.8% of those considered to be normal, as opposed to 10%.
- There is a Spearman correlation r = 0.70 between confidence and IoU for all DEV patches.
- The reliability bins are informative: 77–100% of patches are truly defective while the remaining 15–18% have a lower confidence value.

2. Challenges Faced

1. Near chance localization is the pixel level localization. Pooled AP of 0.035-0.080 is higher than 0.033, which is the chance level, so that the deviation maps fail to detect defects. With the calibrated threshold the F1 is only 0.017 - 0.025, and region growing improves it marginally (0.022 - 0.025), with an increase in the over-segmentation from 1.8× to 2.4×.
2. Low sensitivity at a low false-alarm rate. 25.7% of defect patches are caught by the 5% FPR threshold. Otsu is able to catch all of the objects (recall 0.83), but it labels all of the patches normally as "Q" and it over-segments 13×.
3. The selection of pipelines is fluctuating. There are discrepancies between the three metrics. Adaptive has the best AP and AUROC (0.080 / 0.797) but the lowest Macro AP (0.084), while fixed multi-scale wins Macro AP (0.209) on an AUROC of only 0.552. Fabrics with few patches (less than 20) are driving Macro AP (fabric 06 has 5 patches; fabric 01 has 8 patches; fabric 05 has 0 patches). Therefore, the "carried-forward" pipeline is poorly supported.
4. Confidence will determine patches, not mask quality. The confidence does not correlate with IoU within the patches (Spearman −0.057). It simply means that a defect may or may not exist, not how well it was segmented.

5. Raw-strip demo failed. No detection on three DEV strips (fabric 03); 6 detection with IoU 0 on fabric 01 (top confidence 0.98); 4 detection with IoU 0.003 on fabric 02. The stitching introduces false positives and the 16 px border margin means that a thin band around the strip edges on the sides is not scored.
6. There are no rules for weave periods. All the fabrics have a period of about 4–6 px, which is why the same scales (32/64/128) are selected for each of them by the adaptive rule. The only thing that differs is the smoothing and per-fabric reference. There are no DEV defect patches for Fabric 05, and therefore it cannot be evaluated.
7. Small, imbalanced data. Fabric 03 is the most popular patch in DEV (85 out of 152). The numbers of patches per fabric are noisy in some fabrics, as there are fewer than 10 patches per fabric in some of them.
8. Protocol differences. These outcomes are generated with a distinct protocol as compared with earlier experiments (calibrated thresholds, per-fabric references, interior scoring), therefore are not easily compared till all methods are rerun through this protocol.

3. Work To Be Done

The next step is a single equitable comparison of all techniques. The plan below will further expand the DEV and NTEST protocol in this notebook, which was already expanded for all methods (calibrated thresholds, per-fabric references, interior scoring). All the other splits are left as is until the final comparison.

1. Each evaluation script has one results table.

The same 16 px interior scoring applies: margin is not scored.Calibrated threshold, patches: same for each method.

Store the results per image (method, fabric, patch, defect size, pixel counts, IoU, precision, recall, F1, detected flag, false regions, confidence, peak score, etc.) to a file for analysis and plotting without re-running the methods.

2. Measures to put in place

Pixel-level: Mean per patch and mean values of IoU, precision, and recall against the ground-truth masks.

Map quality (not through the threshold, but) average precision and AUROC on the deviation map (explain why - part done for the three Fourier pipelines; extend to the baselines).

Defect-level: If there is a predicted region that contains a defect, then the defect is counted as detected. Report the number of detected defects and false region/ image, divided by size (small, medium, large).

Image-level: obtain detection decision of each patch as defective or defect-free. From the confidence score, report precision, recall, F1, false-alarm rate and ROC/AUROC.

3. Evaluation data

Don't use the 10-patch sample of defects, but use all defect-containing patches of the DEV split (after margin filtering: 152) along with all defect-free patches of the NTEST split (held out: 280) as false alarms.

4. Methods to compare

Re-run the protocol using the baseline descriptors (LBP, GLCM and Gabor) with the same patches, calibrated thresholds and interior scoring.

Compare them to the fixed single scale, fixed multi scale and adaptive multi scale Fourier spectral residual pipelines.

Note: The baseline code is not included in this notebook, and needs to be ported into the shared script.

5. Re-check the confidence score

Do calibration check on patches with no defect (observed vs expected share above the confidence level).

Re-test for whether confidence is correlated with segmentation accuracy (larger set and all methods). There is currently no patch quality component in detected patches (Spearman 0.70 with IoU overall).

6.Deliverable

A metrics table (pixel-level, defect-level and image-level) for the fixed and adaptive pipelines and the baselines, as well as a saved per-image results file.